

My approach was to transform Shopwise into a **production-ready, local-first voice shopping platform** while preserving its existing visual design and core functionality. We first audited the application architecture, state flow, NLP pipeline, browser APIs, and component dependencies to identify duplicated logic, stale state, hydration risks, and disconnected demo data. We then centralized product normalization, category mapping, quantity/unit parsing, and intent handling into reusable domain modules.

The Zustand architecture was strengthened with persistent local-device storage, synchronized shopping sessions, duplicate-aware quantity merging, multi-action undo, pantry management, purchase history, and safe data reset/export. The voice pipeline was hardened using browser-native Speech Recognition and Speech Synthesis with explicit lifecycle management, permission handling, language support, timeout protection, and deterministic command responses.

Recommendations and insights were connected to real purchase history and pantry data rather than demo values. UI pages were wired to real application state, while accessibility, responsive behavior, hydration safety, error boundaries, SEO, and browser API handling were improved.

Finally, the application was validated through **138 automated tests**, TypeScript checks, ESLint, and production builds to ensure the major shopping, voice, pantry, persistence, and recommendation flows work reliably.